

DR. ROBERT ADRIEL MOSTOGHIU PAUN

Data Scientist – Computational Astrophysicist

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EXECUTIVE SUMMARY

I am a **PhD-qualified Data Scientist and Computational Astrophysicist** with 8+ years of experience engineering large-scale predictive models and automated ETL pipelines. Currently, at the ARC Dark Matter Centre of Excellence, I lead international research projects optimising high-fidelity simulations for complex terabyte-scale datasets, achieving a **40% accuracy increase** in HPC environments and a **98% reduction in engineering cycle times** through AI orchestration. I possess deep proficiency in Python and C/C++, statistical inference, and numerical analysis, with hands-on experience developing AI solutions in scikit-learn and PyTorch.

WORK EXPERIENCE

Postdoctoral Research Associate

ARC Centre for Dark Matter and Particle Physics, Swinburne University

February 2022 – Present

Melbourne, VIC, Australia

- **Numerical System Optimisation:** Engineered custom extensions for the GIZMO computational framework (GIZMO-PBHEF) to model complex system dynamics in distributed High Performance Computing (HPC) environments. Developed and validated novel simulation techniques that achieved a **40% improvement** in model accuracy by identifying and correcting systemic errors within terabyte-scale predictive models.
- **Data Pipeline Engineering:** Architected end-to-end Python and SQL Extract, Transform, Load (ETL) workflows to transform **terabyte-scale raw simulation data** into structured, high-fidelity formats for **distributed HPC processing**.
- **Advanced Anomaly Detection:** Leveraged data science techniques such as A/B testing, feature engineering, statistical thresholds, and population-level diagnostics to identify anomalies and distinguish **genuine signals from computational noise**, ensuring high-fidelity data for decision-making.
- **AI Orchestration:** Engineered the "SAGE Tree Converter", an AI-assisted framework leveraging foundational LLMs via CLI to automate the conversion of complex hierarchical data. Implemented an adapter design pattern and persistent knowledge base logic to transform a 1–3 month manual engineering cycle into a single-day process (**98% reduction in cycle time**). This architecture enables autonomous execution, optimising token usage and ensuring scalable interoperability for non-standard simulation outputs.
- **Operational Infrastructure:** Spearheaded the migration of the "The Three Hundred" **100-member international consortium** to a scalable communication and data-sharing hub. Architected the platform's technical framework and governance, replacing high-cost commercial enterprise tools with a cost-effective solution that streamlined global technical operations and remains the primary collaboration hub.
- **Strategic Governance & Operations:** Appointed to the Swinburne Time Allocation Committee (STACK) to evaluate and rank research proposals, ensuring high-value telescope resources are awarded to projects with the highest scientific ROI. Co-manage remote observing infrastructure and technical support to ensure 24/7 operational readiness for a **multi-million dollar** international research partnership.
- **Stakeholder Engagement & Leadership:** Supervised undergraduate and postgraduate researchers, and delivered **weekly performance visualisations** to internal stakeholders to ensure alignment on complex technical milestones.

PhD Researcher

Universidad Autónoma de Madrid

September 2016 – November 2020

Madrid, Spain

- **Data Standardisation:** Developed Python-based workflows (NumPy/Pandas) to integrate and standardise terabyte-scale datasets distributed across **multiple international HPC facilities**, enabling large-scale comparative analysis for the "The Three Hundred" consortium.
- **Statistical Benchmarking:** Executed rigorous statistical benchmarking of multiple modelling architectures to isolate numerical artefacts from physical signals, establishing robust criteria for **"fair comparisons"**.
- **Technical Communication:** Authored multiple peer-reviewed publications and presented complex data findings to **diverse audiences**, ranging from international specialised conferences to non-technical stakeholders, translating high-dimensional simulation results into actionable scaling laws.

EDUCATION

Universidad Autónoma de Madrid

Madrid, Spain

Doctor of Philosophy in Computational Astrophysics

September 2016 – November 2020

- **Thesis:** CLUES from The Three Hundred: A two-way approach to the influence of baryonic physics on galaxy formation
- **Awards:** Graduated with Summa Cum Laude (2020). Awarded a Doctorate Honourable Mention (2022).

Universidad Autónoma de Madrid

Madrid, Spain

Master of Science in Cosmology and Particle Physics

September 2015 – June 2016

- **Awards:** General enrolment funding and academic stipend by the Spanish Ministry of Education

Universidad Autónoma de Madrid

Madrid, Spain

Bachelor's Degree in Physics

September 2011 – June 2015

- **Awards:** General enrolment funding and academic stipend by the Spanish Ministry of Education

PROFESSIONAL DEVELOPMENT

Commercialisation Training Program for Medical Researchers

Melbourne, Australia

AAMRI VIC

September 2025 – December 2025

- Selected for a competitive program to bridge computational astrophysics knowledge with digital health innovation, developing commercialisation pathways for cross-sector projects. Cultivated expertise in **industry-standard investor pitching** and the translation of digital research into high-impact health solutions.

Research Translation & Design Thinking

Geneva, Switzerland

IdeaSquare, CERN

July 2025

- Selected for an intensive program focused on re-framing technical research for real-world applications through training in commercial literacy and visual storytelling. Developed capabilities in creative problem solving and the translation of **complex technical knowledge for non-specialised audiences**.

SKILLS

Programming & Databases: Python (NumPy, SciPy, Pandas, Polars, Seaborn, Matplotlib), Jupyter, SQL (PostgreSQL, MySQL), R, C/C++.

Data Science & Modelling: Numerical & Predictive Modelling, Machine Learning (scikit-learn, PyTorch), Generative AI, AI Orchestration (CLIs), Bayesian Inference, CI/CD, Large-Scale Data Analysis, Data Visualisation

Developer Tools: High-Performance Computing (HPC), Cloud Services (Azure Databricks), Dashboards (Power BI, Excel), Git, Docker, Linux Shell Script, SLURM, LaTeX

Professional: Project Management, Stakeholder Communication, Cross-Functional Leadership, Research & Problem Solving

Languages: English (fluent), Spanish (native), Romanian (native)